

Module 1: Understanding the Incident Response

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Module Learning Outcomes

- Explain Incident Response (IR)
- Describe IR lifecycle & frameworks
- Develop Incident Response Plan
- Understand playbooks & crisis communication
- Identify IR team roles

Introduction to Incident Response

- Cyber incidents are inevitable
- Prevention alone is insufficient
- Focus on detection, response & recovery
- Minimize damage and downtime

Definition of a Security Incident

- NIST: Violation of security policy
- Examples:
 - Malware
 - Data breach
 - Unauthorized access
 - DoS attacks

Incident vs Event

- Event: Observable activity
- Incident: Confirmed security issue
- Not all events are incidents

Importance of Incident Response

- Reduces financial loss
- Ensures business continuity
- Protects reputation
- Supports legal compliance

IR Lifecycle Overview

- Preparation
- Detection & Analysis
- Containment
- Eradication
- Recovery
- Lessons Learned

Phase 1: Preparation

- Policies & procedures
- Tools & training
- Asset identification
- Playbooks

Preparation Activities

- IR Plan development
- Contact lists
- Legal readiness
- Awareness training

Detection & Identification

- Monitoring systems
- Identify IoCs
- Correlate alerts
- Confirm incidents

Detection Tools

- SIEM
- IDS/IPS
- EDR
- Logs
- User reports

Incident Analysis

- Scope & impact
- Affected assets
- Attack type
- Severity

Containment Phase

- Limit damage
- Prevent spread
- Preserve evidence

Containment Techniques

- Network isolation
- Block IPs
- Disable accounts

Eradication Phase

- Remove malware
- Root cause analysis
- Patch systems

Recovery Phase

- Restore backups
- Validate systems
- Monitor closely

Lessons Learned

- Post-incident review
- Documentation
- Update policies

Incident Response Framework

- Structured approach
- Repeatable process
- Aligns people & technology

NIST IR Framework

- • Preparation
- • Detection & Analysis
- • Containment, Eradication, Recovery
- • Post-Incident Activity

SANS IR Framework

- • Preparation
- • Identification
- • Containment
- • Eradication
- • Recovery
- • Lessons Learned

ISO/IEC 27035

- • Incident management standard
- • Governance focus
- • ISO 27001 aligned

Selecting an IR Framework

- Organization size
- Compliance needs
- Risk profile

Incident Response Plan (IRP)

- Documented procedures
- Guides response
- Ensures coordination

Objectives of IRP

- • Minimize impact
- • Reduce response time
- • Legal compliance

IR Plan Components

- • Definitions
- • Roles
- • Communication
- • Escalation

Incident Classification

- • Type
- • Severity
- • Business impact

Legal & Compliance

- • Evidence handling
- • Chain of custody
- • Breach notification

Incident Response Playbooks

- Step-by-step guides
- Incident-specific
- Manual or automated

IR Plan vs Playbook

- IR Plan: Strategic
- Playbook: Tactical

Playbook Components

- Trigger
- Actions
- Escalation
- Recovery

Malware Playbook

- • Detect
- • Isolate
- • Remove malware
- • Restore

Phishing Playbook

- • Identify email
- • Quarantine
- • Reset credentials

DDoS Playbook

- • Detect traffic
- • Engage ISP
- • Rate limiting

Testing IR

- • Validates readiness
- • Finds gaps

Types of IR Testing

- • Tabletop
- • Simulations
- • Red/Blue team

Tabletop Exercises

- • Scenario
- • Role play
- • Debrief

IR Metrics

- • MTTD
- • MTTR
- • Containment time

Continuous Improvement

- • Update plans
- • Revise playbooks

Incident Response Team

- • Cross-functional team

IRT Roles

- • Manager
- • Analysts
- • Forensics
- • Legal

Skills for IRT

- • Technical
- • Analytical
- • Communication

Internal vs External IRT

- Internal: Context
- External: Expertise

Incident Communication

- Internal updates
- Executive briefings

Escalation Procedures

- • Severity-based
- • Timely decisions

Crisis Communication

- • Manage major incidents
- • Avoid misinformation

Crisis Stakeholders

- • Management
- • Customers
- • Regulators
- • Media

Crisis Communication Plan

- • Approved messages
- • Spokesperson

Social Media Handling

- • Monitor sentiment
- • Be transparent

Post-Incident Review

- • Technical review
- • Communication review

Module Summary

- • IR is essential
- • Frameworks guide response
- • Continuous improvement